

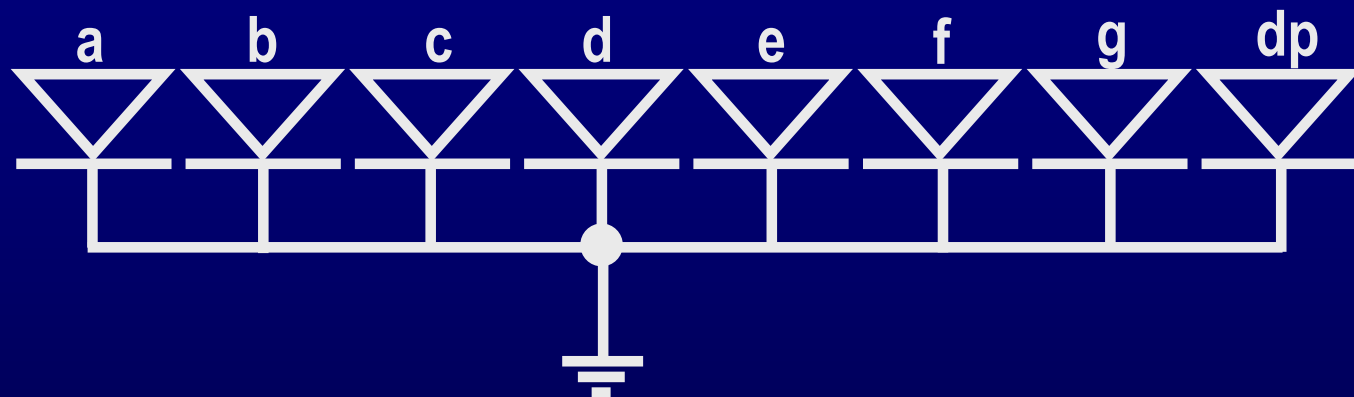
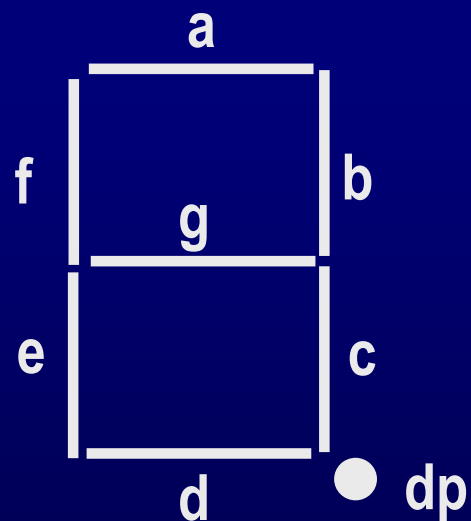
实验任务

7段数码管控制器的设计和实现。

实验要求如下：

1. 7段显示编码模块的设计。
2. 使用1个数码管显示4位2进制数。
3. 时钟分频器的设计。
4. 六个数码管的扫描显示。

7段数码管



7段数码管的显示模块

```
module SevenSegs(
    input  [3:0]      Digit ,
    input             EnableSegs ,
    output reg [6:0] Seg ) ;
    reg [6:0] A_G ;
    always @ (Digit or EnableSegs ) begin
        if (EnableSegs)
            case ( Digit )
// Segment patterns  abcdefg
                0:  A_G = 7'b1111110;  // 0
                1:  A_G = 7'b0110000;  // 1
                2:  A_G = 7'b1101101;  // 2
                3:  A_G = 7'b1111001;  // 3
                4:  A_G = 7'b0110011;  // 4
                5:  A_G = 7'b1011011;  // 5
                6:  A_G = 7'b1011111;  // 6
                7:  A_G = 7'b1110000;  // 7
                8:  A_G = 7'b1111111;  // 8
                9:  A_G = 7'b1110011;  // 9 (no 'tail')
                10: A_G = 7'b1110111;  // A
                11: A_G = 7'b0011111;  // b
                12: A_G = 7'b1001110;  // C
                13: A_G = 7'b0111101;  // d
                14: A_G = 7'b1001111;  // E
                15: A_G = 7'b1000111;  // F
                default: A_G = 7'bx;
            endcase
        else
            A_G = 7'b0;
            Seg = {A_G[0],A_G[1],A_G[2],A_G[3],A_G[4],A_G[5],A_G[6]} ;
        end
    end
endmodule
```

显示四位二进制数

```
module OneSevenSegs_Anvyl(  
    input  [3:0] Switch ,  
    input          Btn0 ,  
    output [6:0] Segs ,  
    output          En    );  
  
    assign En = ~Btn0 ; // Btn0 means for Disable  
  
    SevenSegs M1_SevenSegs( .Digit(Switch) ,  
        .EnableSegs(En) , .Seg(Segs) );  
  
endmodule
```

约束文件

#Switch

NET Switch[0] LOC= V5;

NET Switch[1] LOC= U4;

NET Switch[2] LOC= V3;

NET Switch[3] LOC= P4;

#Segment digisplay Enable

NET EN LOC = M17; # 7-seg Enable 最右1个

7-seg display

NET Segs[0] LOC=AA21; # 7-seg CA

NET Segs[1] LOC=AA22; # 7-seg CB

NET Segs[2] LOC=Y22; # 7-seg CC

NET Segs[3] LOC=N15; # 7-seg CD

NET Segs[4] LOC=AB19; # 7-seg CE

NET Segs[5] LOC=P20; # 7-seg CF

NET Segs[6] LOC=Y21; # 7-seg CG

#Button

Net Btn0 Loc= E6;

时钟分频器

```
module ClockDivider(  
    input      Reset ,  
    input      Clock ,  
    output reg DividedClock    );  
    parameter COUNTER = 32'hC350; // 50 000  
    reg [32:0] clock_count;  
    always @(posedge Clock)  
        if (Reset == 1'b1)  
            clock_count <= {32{1'b0}};  
        else  
            begin  
                if ( clock_count == COUNTER-1 ) // CLK为100M时,  
dividedClk的周期 1ms , 高低各 0.5ms  
                    begin  
                        DividedClock <= ~DividedClock;  
                        clock_count <= {32{1'b0}};  
                    end  
                else  
                    clock_count <= clock_count + 1;  
                end  
            end  
endmodule
```

六个数码管的扫描显示模块

```
module SevenSegsController(  
    input      Reset ,  
    input      Clock ,  
    input  [5:0] DisplayEnables ,  
    input  [23:0] Data ,  
    output [7:0] SevenSegsAndPoint,  
    output [5:0] ShowOneofSix  
);  
    reg [2:0]    select_display;  
    wire [3:0]   four_digits;  
    always @(posedge Clock)  
        if (Reset == 1'b1)  
            select_display <= 3'b000;  
        else  
            begin  
                if (select_display == 5 )  
                    select_display <= 3'b000;  
                else  
                    select_display <= select_display + 1;  
            end  
end
```

```

        assign four_digits = (select_display[2:0] == 3'b000) ? Data[3:0] :
(select_display[2:0] == 3'b001) ? Data[7:4] :
(select_display[2:0] == 3'b010) ? Data[11:8] :
(select_display[2:0] == 3'b011) ? Data[15:12] :
(select_display[2:0] == 3'b100) ? Data[19:16] :
                                                                    Data[23:20];
    assign ShowOneofSix =
        (select_display[2:0] == 3'b000 & DisplayEnables[0] == 1'b1) ?
6'b000001 :
        (select_display[2:0] == 3'b001 & DisplayEnables[1] == 1'b1) ?
6'b000010 :
        (select_display[2:0] == 3'b010 & DisplayEnables[2] == 1'b1) ?
6'b000100 :
        (select_display[2:0] == 3'b011 & DisplayEnables[3] == 1'b1) ?
6'b001000 :
        (select_display[2:0] == 3'b100 & DisplayEnables[4] == 1'b1) ?
6'b010000 :
        (select_display[2:0] == 3'b101 & DisplayEnables[5] == 1'b1) ?
6'b100000 :
        6'b000000;
    SevenSegs M1_SevenSegs(.Digit(four_digits) ,
.EnableSegs(DisplayEnables[select_display]) , .Seg(SevenSegsAndPoint[6:0])) ;

    assign SevenSegsAndPoint[7] = 1'b0;
endmodule

```


顶层模块

```
module SixSevenSegs_Anvyl(  
    input          Clock,  
    input  [5:0]    Switch,  
    output [5:0]    En ,  
    output [7:0]    Segs          );  
    wire          divided_clock;  
    wire  [23:0]   data ;  
    assign data = 24'h543210 ;  
    ClockDivider M1_ClockDivider(.Reset(1'b0),  
.Clock(Clock), .DividedClock(divided_clock));  
    SevenSegsController M2_SevenSegsController(  
        .Reset(1'b0), .Clock(divided_clock),  
.DisplayEnables(Switch),  
.Data(data), .SevenSegsAndPoint(Segs), .ShowOneofSix(En));  
endmodule
```

实际测试

